



JSP IV

1. EL
2. JSTL
3. Custom tags

- The main purpose of it to simplify the process of accessing data from bean properties and from implicit objects.
- Syntax of EL:
 - `${ expr.expr }` or `${ expr[expr] }`
 - `expr` gets evaluated at runtime
 - In general `dot(.)` operator used for accessing bean property or map key and `bracket([])` operator used when bean is of any type of collection.
- Advantages of EL
 - compact syntax
 - allows to access nested properties
 - better job of handling null values
 - EL has it's own set of implicit objects
 - No need to specify the scope of the bean.

- Disadvantages of EL
 - Can not create a JavaBean
 - No way to update properties
 - No provision to iterate over bean of type collection.
- Different ways to use EL
 - `${1+2+3}`
 - `<p> The current date is ${currentDate}</p>`
 - `<p> City : ${user.address.city} </p>`
 - `Hello, ${cookie.name.value}`
 - `Please contact ${iParam.adminemail}`
- Use `[ ]` operator to work with arrays and lists
- EL Operators: supports arithmetic, relational and logical operators
- Preventing EL Evaluation

- JSP Standard Tag Library(JSTL) is a standard library of readymade tags.
- Several tags in JSTL can remove scriptlet code from a JSP page.
- JSTL is divided into 5 groups:
 1. JSTL Core: support some basic scripting task.
 2. JSTL Formatting: to format text, date, number
 3. JSTL sql: to perform database operations
 4. JSTL XML: support for XML processing
 5. JSTL functions: support for string manipulation
- JSTL core tags:
 - `<c:if .../>`: simple conditioning logic
 - `<c:forEach .../>`: provides way to iterate over a bean.
 - `<c:redirect .../>`: helps to redirect request
 - and many more

- Examples
- Steps for using JSTL
 1. Add the JSTL jar files in WEB-INF/lib folder of deployed application
 2. Add taglib directive to specify URI and prefix
 3. Use required tags

- A custom tag library is a set of custom tags that invoke custom actions in a JavaServer Pages (JSP) file.
- Benefits of tag libraries
 - Simplify complex actions
 - Help separate presentation from implementation
- Syntax

```
<prefix:tag name attr1=value1.... attr = value>
    body code if needed
</prefix:tagname>
```
- Steps for implementing custom tags
 1. Tag handler class
 2. Tag descriptor file (TLD) file
 3. Add taglib directive on JSP page and use the tag. In taglib directive, specify URI as path of tld file